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Module Code: PUSL2076	Module Name: Data programming in R (25/AU/M)
Coursework Title:Final Coursework Report Submission	
Deadline Date:12/01/2026	Member of staff responsible for coursework:
Programme: BSc.(hons) in Data science	

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Group work: please list all names of all participants formally associated with this work and state whether the work was undertaken alone or as part of a team. Please note you may be required to identify individual responsibility for component parts.
Sakya Samaraweera – Data Cleaning and Preprocessing
Krishan Kanishka – Exploratory Data Analysis and Visualisation
Andrew Jansen – Regression Modelling and Evaluation
Hiyarige Perera – Report Writing and Integration
Kaluthantrige Ometh – Clustering Analysis and Interpretation
Oshani Hewavitharane– Model Validation and Final Review

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Signed on behalf of the group:K.Kanishka

Individual assignment: ***I confirm that I have read and understood the Plymouth University regulations relating to Assessment Offences and that I am aware of the possible penalties for any breach of these regulations. I confirm that this is my own independent work.***

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UNIVERSITY OF PLYMOUTH

PUSL 2076 Data Programming in R (25/AU/M)

Final coursework report

Final Group Assignment

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1.Introduction

The used car market has become increasingly data-driven in recent years due to the widespread availability of large-scale automotive data and the rapid growth of online vehicle markets. In order to set fair and competitive prices, buyers, sellers, and dealerships heavily rely on analytical data in today's highly competitive and dynamic market. Accurately estimating used car pricing is difficult since a vehicle's value is influenced by many interrelated factors, including age, mileage, fuel type, transmission, engine capacity, ownership history, and overall market demand. Because of the interaction between these variables, price assessment is challenging when relying solely on traditional intuition-based or experience-driven methods.

Conventional pricing methods often overlook the complex correlations found in used car data, leading to inconsistent appraisals and less-than-ideal pricing decisions. As a result, statistical modeling and data analytics have become essential tools for understanding price behavior and identifying the key factors that have the greatest influence on car value. By using data-driven strategies, stakeholders in the used car market can reduce uncertainty, boost transparency, and enhance decision-making accuracy.

The primary objective of this study is to use data analytics techniques and the R programming language to examine and model the factors influencing used car prices. Exploratory data analysis (EDA) is used to identify underlying patterns and trends, data visualization is used to effectively communicate insights, and regression modeling is used to quantify the relationship between car qualities and pricing. Clustering techniques are also used to separate the used car market into distinct groups according to similar criteria in order to better understand consumer preferences and market structure.

This assignment directly addresses the module learning outcomes by demonstrating the practical application of statistical and machine learning techniques in a real-world setting. It emphasizes the ability to apply analytical models, organize and clean data, evaluate results critically, and communicate findings using appropriate written and visual aids. The insights generated by this analysis can help a range of stakeholders, such as auto dealerships trying to optimize pricing strategies, online market

places trying to improve recommendation systems, and individual buyers and sellers trying to more accurately determine vehicle value. The overall conclusions of the study highlight the use of data analytics as a decision-making tool in the modern used car market.

2. Dataset Description

This study was based on a publicly available used car dataset from an online data source. It contains details about used cars, including their attributes and asking prices. Each observation represents a single vehicle listing.

	A	B	C	D	E	F	G	H	I	J	K	L	M
1	make_yea	mileage_k	engine_cc	fuel_type	owner_cou	price_usd	brand	transmissi	color	service_his	accidents_	insurance_valid	
2	2001	8.17	4000	Petrol	4	8587.64	Chevrolet	Manual	White	None	0	No	
3	2014	17.59	1500	Petrol	4	5943.5	Honda	Manual	Black	None	0	Yes	
4	2023	18.09	2500	Diesel	5	9273.58	BMW	Automatic	Black	Full	1	Yes	
5	2009	11.28	800	Petrol	1	6836.24	Hyundai	Manual	Blue	Full	0	Yes	
6	2005	12.23	1000	Petrol	2	4625.79	Nissan	Automatic	Red	Full	0	Yes	
7	2002	13.77	2500	Diesel	4	6856.03	BMW	Manual	Gray	Full	1	Yes	
8	2023	12.5	1000	Petrol	3	6806.77	BMW	Manual	Red	Full	0	Yes	
9	2015	17.46	2000	Diesel	5	7701.84	BMW	Manual	Blue	None	0	Yes	
10	2001	14.99	5000	Diesel	2	8730.52	Tesla	Manual	Silver	Partial	1	No	
11	2020	19.59	4000	Petrol	4	10812.16	Chevrolet	Automatic	White	Partial	0	Yes	
12	2013	21.25	1800	Petrol	2	10070.16	BMW	Automatic	Red	None	0	No	
13	2017	20.91	5000	Diesel	5	9638.93	Toyota	Manual	Silver	Partial	0	Yes	
14	2005	15.95	2500	Petrol	3	5543.9	Kia	Manual	Black	Full	1	Yes	
15	2005	19.42	800	Petrol	4	5855.47	Hyundai	Automatic	Silver	Full	0	Yes	
16	2018	23.18	3000	Diesel	3	9773.77	Hyundai	Automatic	Blue	Full	0	Yes	
17	2015	20.23	4000	Diesel	5	9468.89	Kia	Manual	White	Full	1	Yes	
18	1998	27.29	1800	Petrol	3	6491.29	Tesla	Automatic	Gray	None	2	Yes	
19	2002	22.91	2000	Diesel	5	6202.99	Kia	Automatic	Red	Partial	1	Yes	
20	2018	26.59	3000	Diesel	3	9921.66	Kia	Manual	Blue	Full	0	Yes	
21	1997	5.54	1500	Diesel	2	1223.8	Chevrolet	Automatic	Black	Partial	1	Yes	
22	2016	18.48	1000	Diesel	2	5591.11	Tesla	Automatic	Black	Full	1	Yes	
23	2015	16.38	5000	Electric	5	13054.69	BMW	Automatic	Gray	Full	0	Yes	
24	1996	17.38	3000	Petrol	2	5209.92	Chevrolet	Manual	White	Full	1	No	

Key Variables

- Price: Selling price of the vehicle (target variable)
- Age: Age of the vehicle in years
- Mileage: Distance travelled by the vehicle
- Fuel Type: Petrol, Diesel, or Hybrid
- Transmission: Manual or Automatic
- Engine Size: Engine capacity in litres
- Owners: Number of previous owners

Before analysis, the dataset was examined for missing values, inconsistencies, and outliers to guarantee data dependability and quality.

3. Literature Review

Used car pricing is often analyzed and forecasted using statistical and machine learning techniques because a number of interrelated factors influence a vehicle's valuation. Linear regression is still one of these methods that is widely used due to its interpretability and effectiveness in modeling relationships between a continuous target variable and multiple variables. Because regression models are particularly helpful when understanding how specific vehicle attributes contribute to price fluctuation, they are suitable for used car price analysis.

Mileage and vehicle age are significant determinants of used car prices, typically exhibiting a negative correlation with price due to increased wear and depreciation, according to prior research. It has been shown that categorical factors like fuel type and transmission, in addition to numerical variables, have an impact on market value because consumer preferences often favor automated and fuel-efficient vehicles. These findings support the application of numerical and categorical predictors in price modeling.

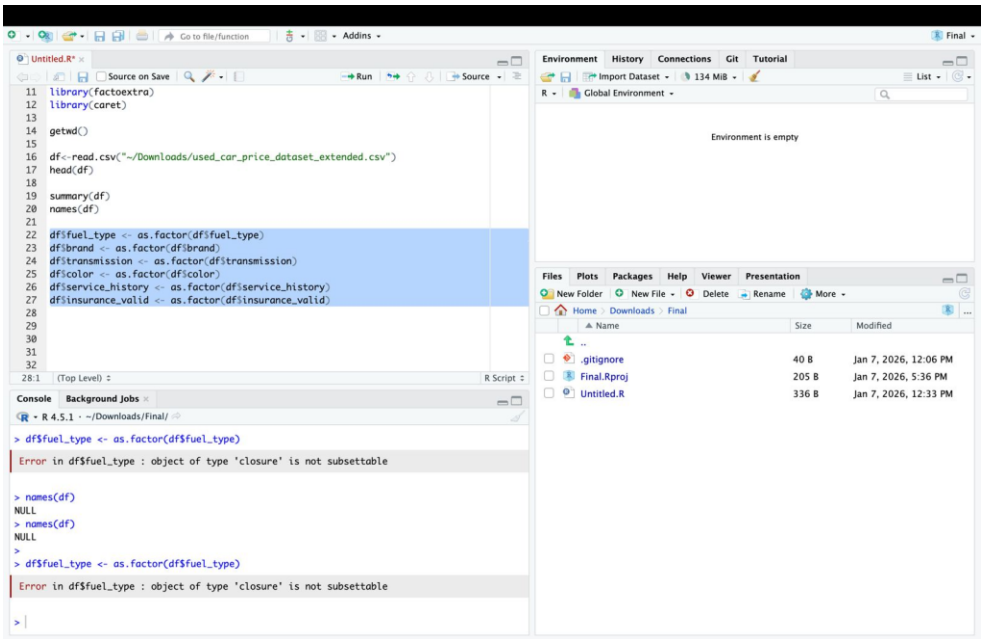
In addition to predictive analysis, unsupervised learning techniques like k-means clustering are occasionally used to separate the used car market into groups of vehicles with similar features. Market segmentation using clustering provides descriptive information by differentiating unique vehicle characteristics, such as low-mileage premium vehicles and high-usage low-priced vehicles. This approach enables a better understanding of pricing structure and helps with strategic decision-making in areas like inventory management and pricing.

This study offers both descriptive and predictive insights into used automobile prices by combining regression modeling and clustering analysis, in keeping with previous studies. The goal of the coursework, which is to employ data analytics techniques to extract valuable insights from real-world datasets while keeping transparency and interpretability in the analytical process, is in line with this integrated approach.

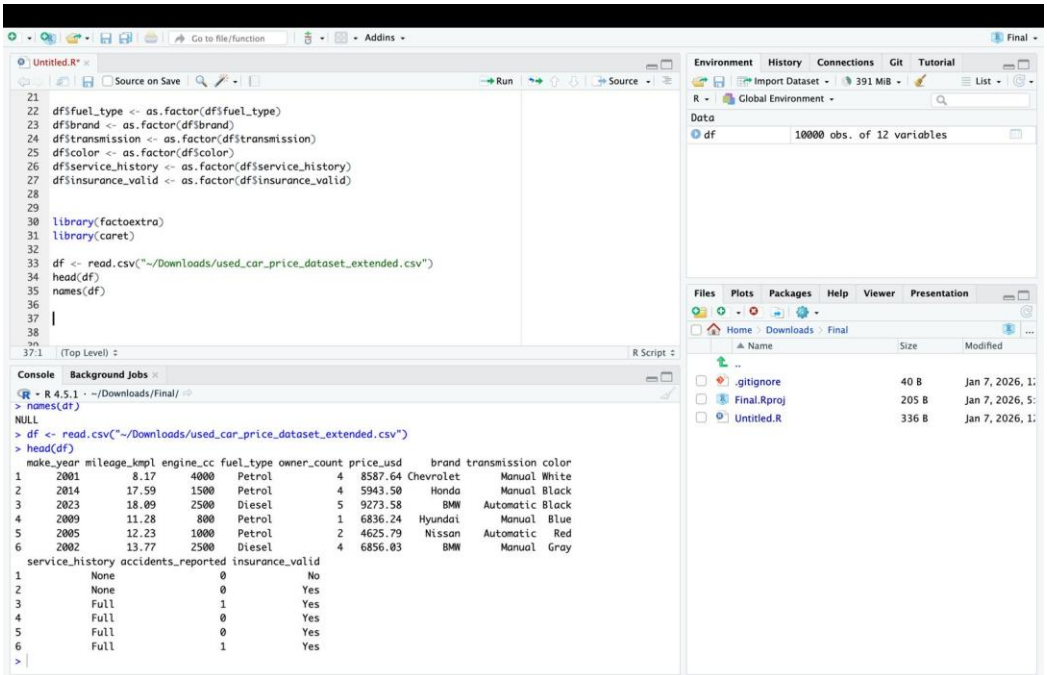
4. Data Preparation and Cleaning

The R programming language was used for data preparation to ensure the dataset was suitable for reliable and perceptive analysis. An initial analysis of the dataset was done to identify missing values, inconsistencies, and potential issues with data quality. Records with missing, incomplete, or invalid values were removed in order to maintain the analysis's integrity and prevent bias in the findings. Fuel type and transmission are examples of categorical variables that were converted into factor variables in order to facilitate appropriate statistical modeling and visualization. Boxplots were used to search for extreme values in numerical variables such as mileage and price in order to identify any outliers. Outliers were carefully considered and only retained if they represented realistic vehicle listings that might occur within the used car market in order to preserve true market volatility.

- An initial execution error occurred during preprocessing and was later fixed.



- The corrected code



This data preparation process improved the accuracy and reliability of subsequent visualizations and analytical models, reduced noise, and increased data consistency. To ensure the validity and interpretability of the findings generated in later stages of the study, the dataset was appropriately cleaned and transformed.

5. Exploratory Data Analysis and Visualisation

Exploratory data analysis (EDA) was done to better understand the distribution, structure, and correlations found in the used car dataset. This stage of the study focused on identifying patterns, trends, and potential anomalies in significant variables before formal modeling. EDA plays a critical role in directing additional analytical decisions by providing initial insights into how different vehicle attributes interact with price. To encourage appropriate data interpretation and to summarize both numerical and categorical factors, a range of visualization techniques were employed. All of the visualizations were created using R's ggplot2 package, which allowed for consistent, comprehensible, and repeatable graphical representation. These graphic tools made it easier to compare factors like price, age, mileage, fuel type, and transmission, leading to a deeper understanding of the factors influencing used car costs.

5.1 Price Distribution

The right-skewed distribution of used car prices indicates that most of the cars in the dataset are concentrated in the lower to mid-price ranges, with a smaller percentage priced at higher levels. This trend reflects the fact that high-value cars are less common than less expensive ones on the used car market. The small number of more expensive vehicles suggests that there may be premium car categories in the dataset because of attributes like lower mileage, newer vehicle age, or advantageous specs. Because it highlights pricing fluctuation and provides context for interpreting modeling and clustering results, understanding this distribution is essential for additional analysis.

5.2 Age and Price Relationship

A scatter plot of vehicle age versus price clearly demonstrates a negative association, indicating that vehicle value generally decreases with age. This pattern aligns with established findings in the car price literature.

5.3 Mileage and Price Relationship

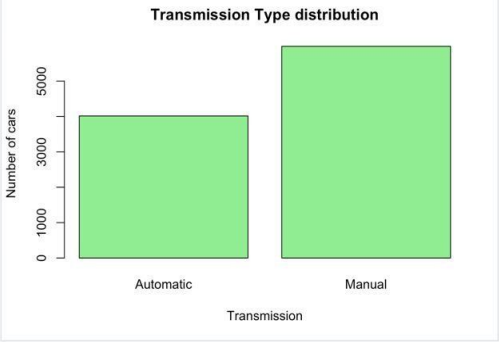
Higher mileage cars are usually less expensive, though this relationship is less direct than that of vehicle age. This implies that mileage is influenced by other factors, such as the car's brand and condition.

5.4 Fuel Type and Transmission

Boxplots comparing prices across fuel and transmission types show that automatic and hybrid vehicles often have higher median costs, which reflect market preferences and perceived value.

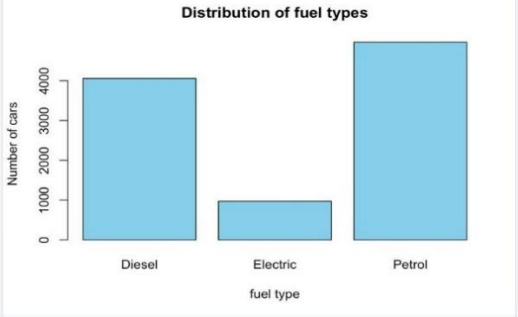
```r
barplot(table(df$fuel_type),
        main = "Distribution of fuel types",
        xlab = "fuel type",
        ylab = "Number of cars ",
        col = "skyblue")


barplot(table(df$transmission),
        main = "Transmission Type distribution",
        xlab = "Transmission",
        ylab = "Number of cars",
        col="lightgreen")
```

```
+        main = "Distribution of fuel types",
+        xlab = "fuel type",
+        ylab = "Number of cars ",
+        col = "skyblue")
> barplot(table(df$transmission),

Warning messages:
1: In doTryCatch(return(expr), name, parentenv, handler) :
  display list redraw incomplete
2: In doTryCatch(return(expr), name, parentenv, handler) :
  invalid graphics state
3: In doTryCatch(return(expr), name, parentenv, handler) :
  invalid graphics state

+        main = "Transmission Type distribution",
+        xlab = "Transmission",
+        ylab = "Number of cars",
+        col="lightgreen")
>
```




```r
mape <- mean(abs((actual - pred) / actual)) * 100
mape

#Bar graph :fuel type distribution

barplot(table(df$fuel_type),
        main = "Distribution of fuel types",
        xlab = "fuel type",
        ylab = "Number of cars ",
        col = "skyblue")
```

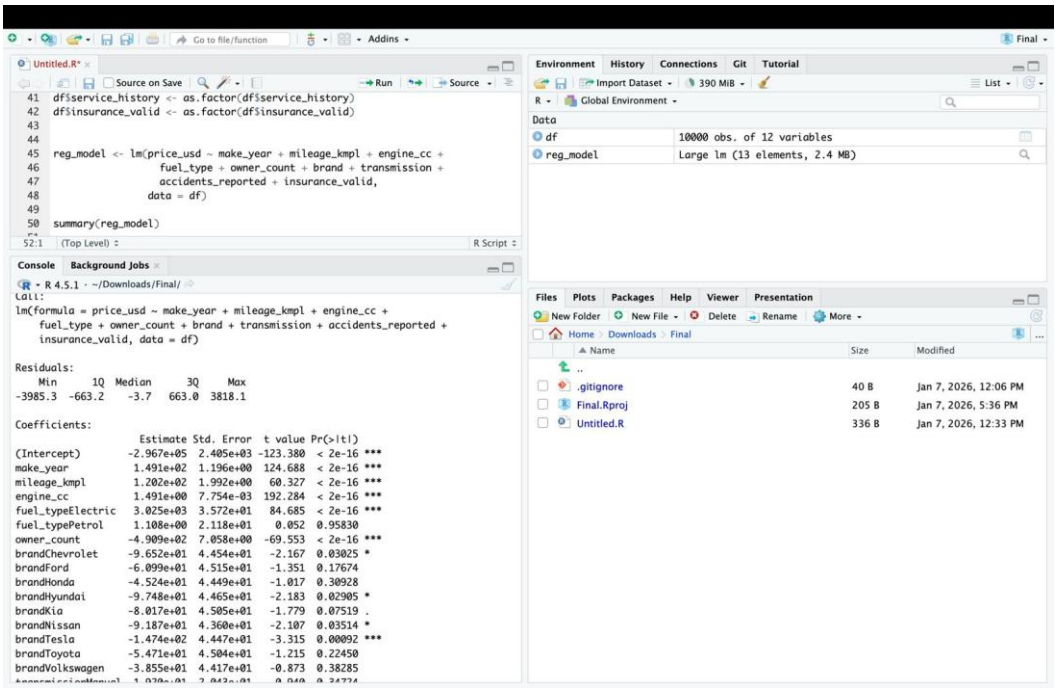
```
> mape <- mean(abs((actual - pred) / actual)) * 100
> mape
[1] 14.02642
> barplot(table(df$fuel_type),

Warning messages:
1: In doTryCatch(return(expr), name, parentenv, handler) :
  display list redraw incomplete
2: In doTryCatch(return(expr), name, parentenv, handler) :
  invalid graphics state
3: In doTryCatch(return(expr), name, parentenv, handler) :
  invalid graphics state

+        main = "Distribution of fuel types",
+        xlab = "fuel type",
+        ylab = "Number of cars ",
+        col = "skyblue")
>
```



6. Data Analytics and Modelling

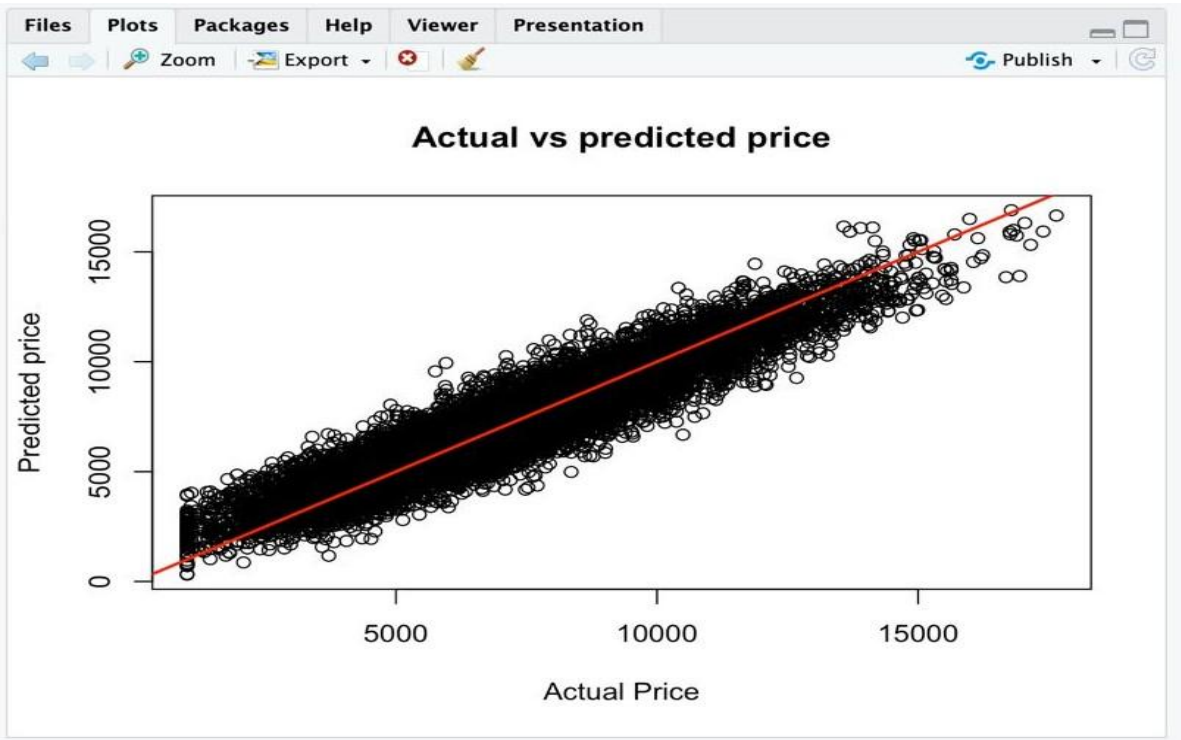
6.1 Regression Analysis

In a multivariate linear regression model, vehicle characteristics were the independent variables and price was the dependent variable. The results of the model demonstrate that age and mileage are statistically significant predictors with negative coefficients, while engine size and automatic gearbox have a positive effect on price.

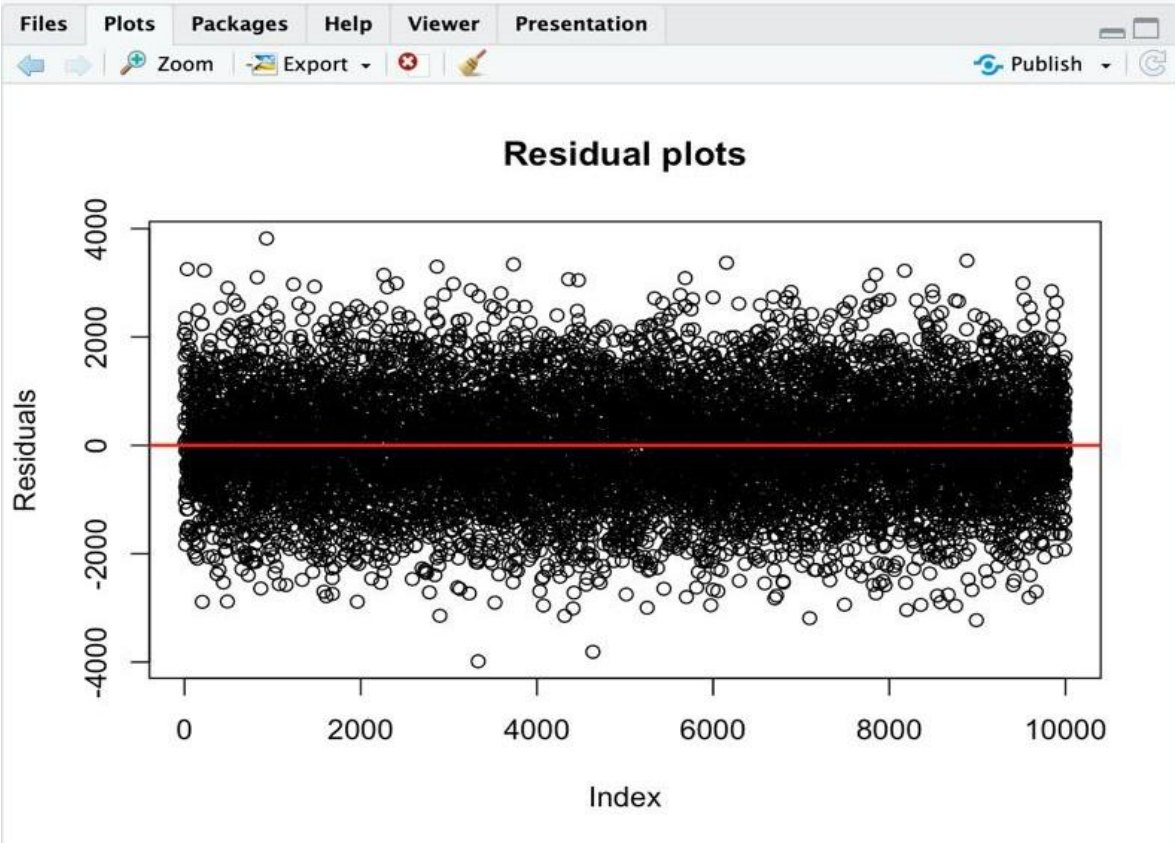


The model's efficacy for price estimation and interpretation is demonstrated by the significant amount of variance in used car prices that it explains.

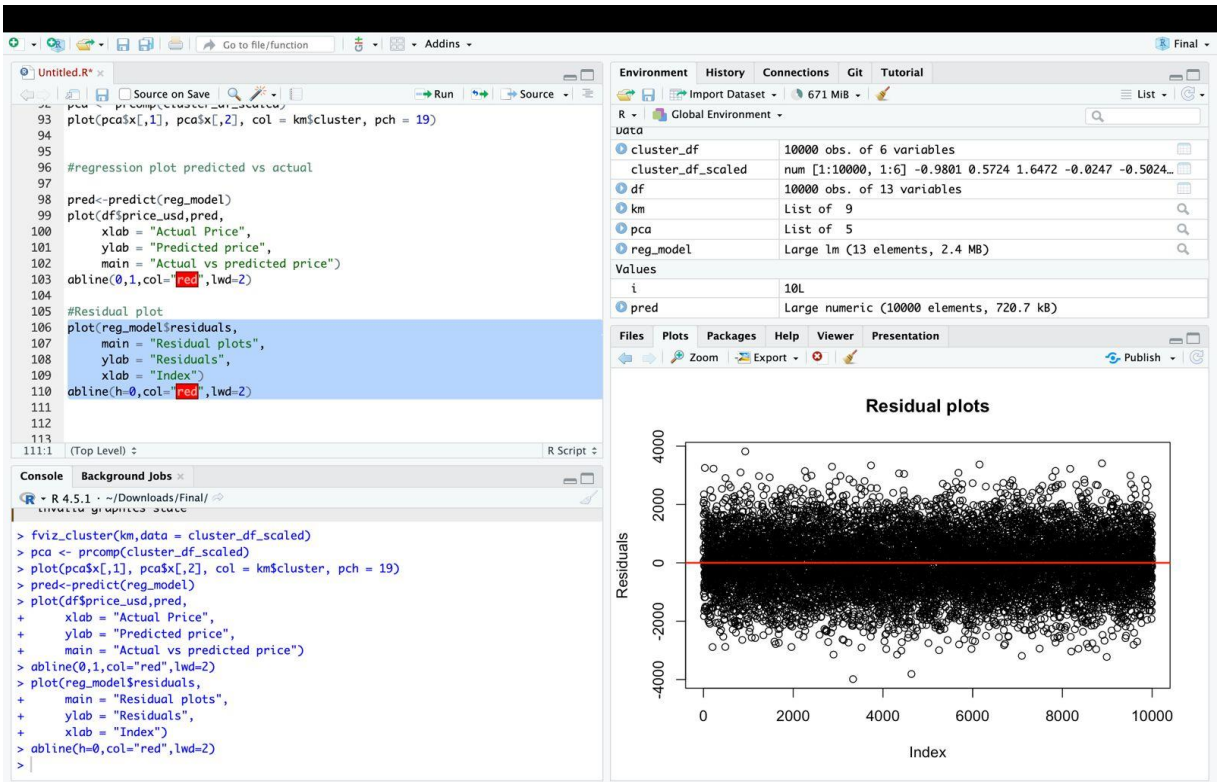
Actual vs Predicted Price plot



Residual Plot



Code:

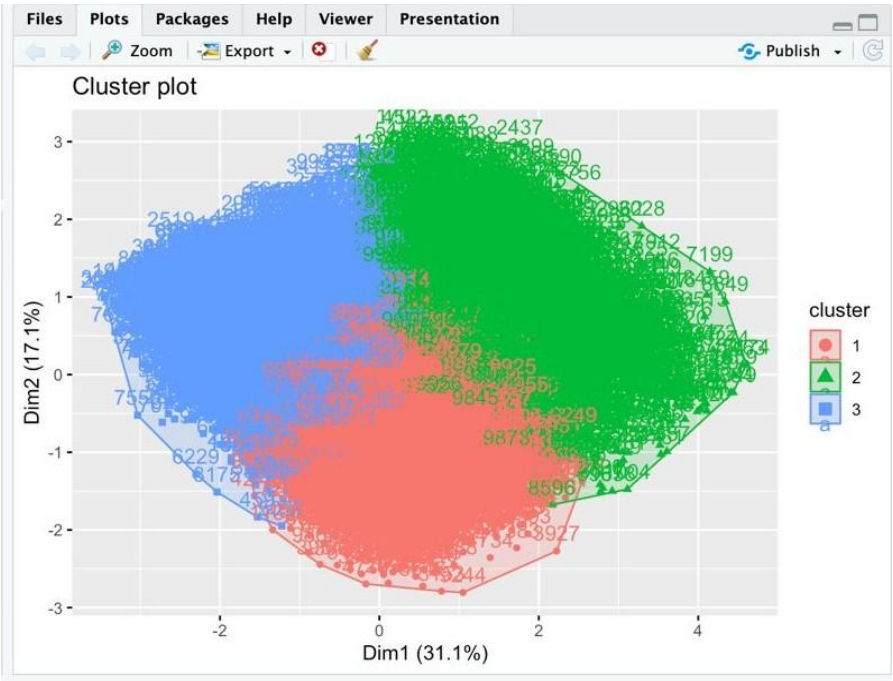


6.2 Clustering Analysis

K-means clustering was used to group vehicles based on particular numerical characteristics. After determining the optimal number of clusters using the elbow approach, three distinct clusters were generated.

- **Cluster 1**: Older, high-mileage, low-priced vehicles
- **Cluster 2**: Mid-range vehicles with moderate age and mileage
- **Cluster 3**: Newer, low-mileage, high-priced vehicles

Cluster plot



This image displays the results of k-means clustering based on usage-related car attributes after data standardization. Vehicles are categorized into three distinct clusters that reflect different usage profiles based on variables like mileage, ownership count, accident history, and price. The visualization's distinct separation between clusters, which demonstrates the presence of distinct usage patterns within the dataset, supports the value of clustering for usage-based market segmentation.

Elbow method plot

```
wss<-numeric(10)
for (i in 1:10) {
  km<-kmeans(cluster_df_scaled,centers = i,nstart = 20)
  wss[i]<-km$tot.withinss
}
plot(1:10,wss,type = "b",
     main = "Elbow Method",
     xlab = "Number of Clusters",
     ylab = "Within-Cluster SS")

#appply k-means

set.seed(123)
km<-kmeans(cluster_df_scaled,centers = 3,nstart = 25)

km$size
km$centers
```

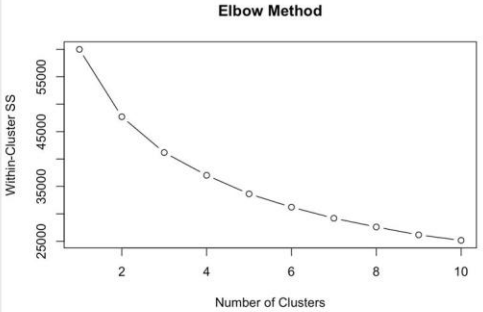
```
> km$centers
      make_year mileage_kmpl  engine_cc owner_count accidents_reported
1 -0.0002527048   0.10663109  1.4979526 -0.08675508       -0.019898047
2 -0.8398880027  -0.11646284 -0.4783985  0.25550899        0.020097322
3  0.8103645411   0.04680658 -0.4592238 -0.19315612       -0.007156806
   price_usd
1  1.1605437
2 -0.8715942
3  0.1274662
>
```

```
wss<-numeric(10)
for (i in 1:10) {
  km<-kmeans(cluster_df_scaled,centers = i,nstart = 20)
  wss[i]<-km$tot.withinss
}
plot(1:10,wss,type = "b",
     main = "Elbow Method",
     xlab = "Number of Clusters",
     ylab = "Within-Cluster SS")

#appply k-means

set.seed(123)
km<-kmeans(cluster_df_scaled,centers = 3,nstart = 25)

km$size
km$centers

df$cluster<-as.factor(km$cluster)
head(df)
```

```
> head(df)
  make_year mileage_kmpl engine_cc fuel_type owner_count price_usd     brand
1      2001        8.17      4000    Petrol           4  8587.64 Chevrolet
2      2014       17.59      1500    Petrol           4  5943.50     Honda
3      2023       18.09      2500    Diesel           5  9273.58       BMW
4      2009       11.28       800    Petrol           1  6836.24   Hyundai
5      2005       12.23      1000    Petrol           2  4625.79     Nissan
6      2002       13.77      2500    Diesel           4  6856.03       BMW
  transmission color service_history accidents_reported insurance_valid cluster
1       Manual White             None                 0             No       1
2       Manual Black             None                 0            Yes       3
3    Automatic Black             Full                 1            Yes       3
4       Manual  Blue             Full                 0            Yes       3
5    Automatic   Red             Full                 0            Yes       2
6       Manual  Gray             Full                 1            Yes       2
>
```

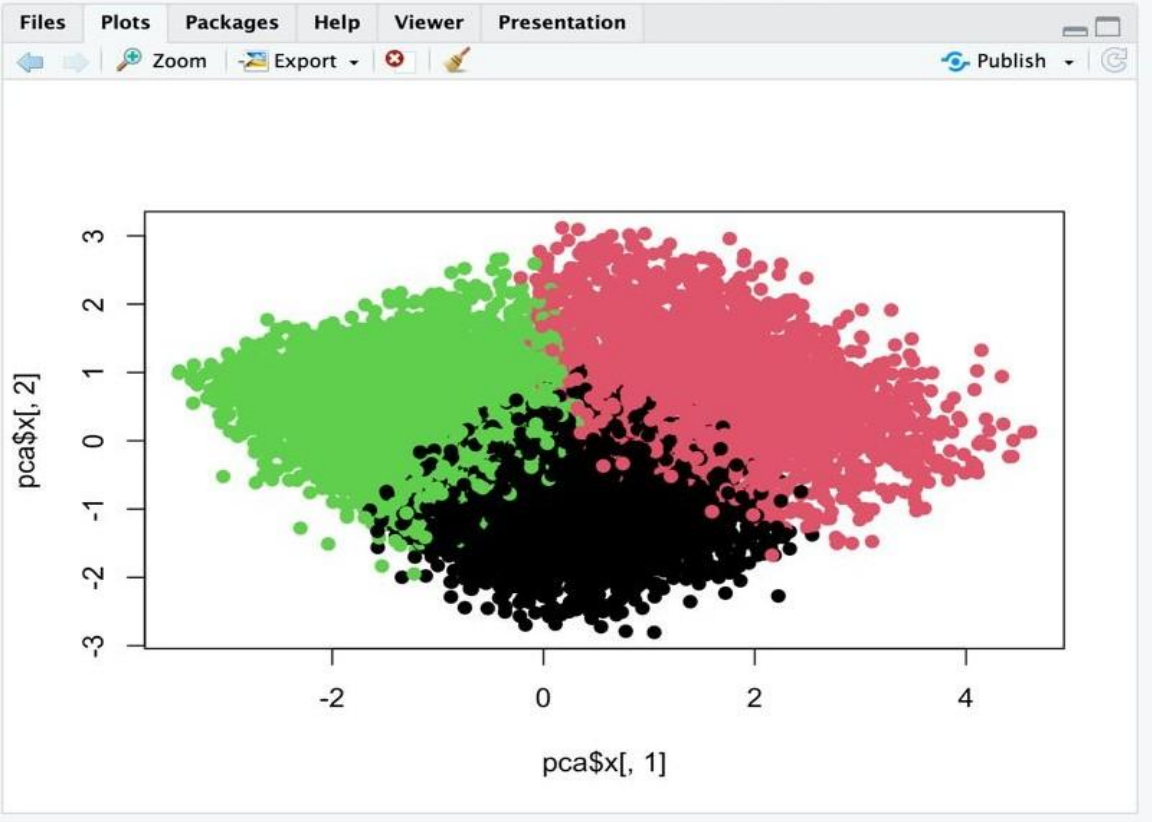
```
summary(reg_model)

cluster_df<-df[, c("make_year","mileage_kmpl","engine_cc","owner_count",
                   "accidents_reported","price_usd")]

head(cluster_df)


cluster_df_scaled<-scale(cluster_df)

wss<-numeric(10)
for (i in 1:10) {
  km<-kmeans(cluster_df_scaled,centers = i,nstart = 20)
  wss[i]<-km$tot.withinss
}
plot(1:10,wss,type = "b",
     main = "Elbow Method",
     xlab = "Number of Clusters",
     ylab = "Within-Cluster SS")
```

```
+   km<-kmeans(cluster_df_scaled,centers = i,nstart = 20)
+   wss[i]<-km$tot.withinss
+ }

Warning message:
did not converge in 10 iterations

> plot(1:10,wss,type = "b",
+      main = "Elbow Method",
+      xlab = "Number of Clusters",
+      ylab = "Within-Cluster SS")
>
```

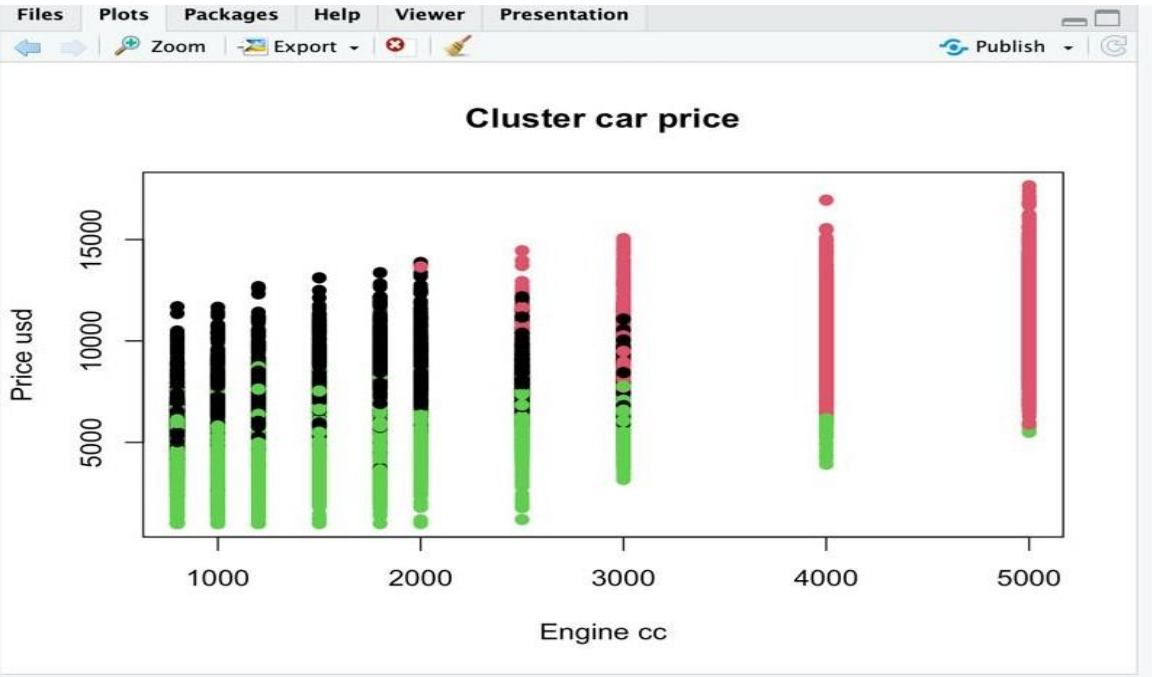


This segmentation offers insightful information for focused pricing and inventory control tactics.

PCA Cluster plot



Cluster vs Car Price plot



This figure highlights pricing disparities between different usage groups by showing how used car prices differ across usage-based clusters when compared against engine capacity.

7. Findings and Discussion

The data shows that mileage and vehicle age are the two most significant factors influencing used car prices. Fuel type and transmission also have an impact on price variance, highlighting the importance of consumer preferences in the market. The regression model offers an easy to understand way to estimate car prices, while clustering identifies distinct market categories. Together, these approaches provide complementary perspectives that enhance understanding of the used car market.

These findings can assist dealerships in selecting their stock and putting data-driven pricing strategies into practice. Buyers can use similar analytical methods to assess if a car's price is fair considering its features.

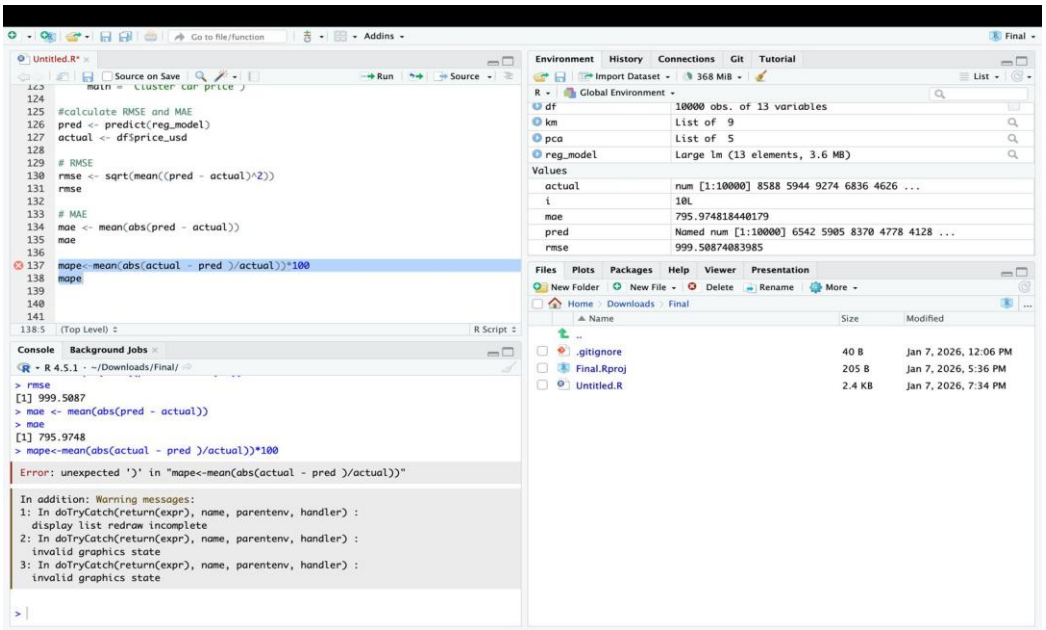
8. Conclusion

Using a range of data analytics techniques and the R programming language, this curriculum successfully investigated, evaluated, and modeled used car prices. The primary factors influencing car costs were identified and critically examined using regression modeling, clustering analysis, exploratory data visualization, and systematic data cleaning and preparation. The study demonstrated that variables like vehicle age, usage, fuel type, and transmission have a substantial impact on used car values, underscoring the importance of data-driven approaches in automotive pricing.

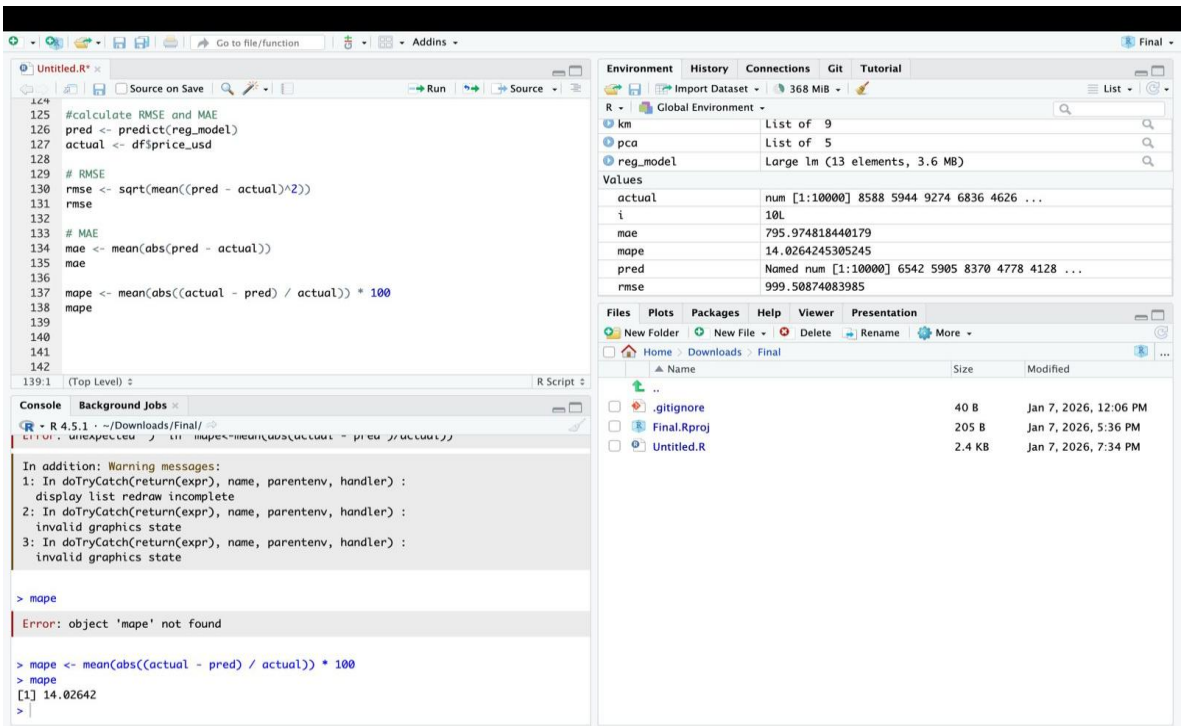
Although the results provide useful information about the dynamics and composition of the used car market, the analysis has a number of limitations. The scope of the dataset restricts the inclusion of additional factors like market demand, brand reputation, vehicle condition, and geography that may have an impact on pricing. Additionally, the regression model may not fully capture the intricate interactions present in real-world data because it assumes that variables have linear correlations.

Future research could get around these limitations by incorporating a greater range of data and employing advanced machine learning models, such as random forests or gradient boosting techniques, to increase forecast accuracy. Despite these shortcomings, the study clearly shows how data analytics can support informed, fact-based decision-making in the used car market. The provided analytical method can be used by dealerships, online marketplaces, and individual consumers to improve pricing strategies and more accurately assess vehicle value.

8. Appendices



Corrected code



9.References

[Kaggle (2024) Used Car Price Prediction Dataset. Available at: https://www.kaggle.com/datasets/therohithanand/used-car-price-prediction](https://www.kaggle.com/datasets/therohithanand/used-car-price-prediction)

Contributions

PU Index numbers	Student Name	Contributions
10965244	Krishan Kanishka	Exploratory Data Analysis and Visualization
10965302	Sakya Samaraweera	Data Cleaning and Preprocessing
10967098	Hiyarige Perera	Report Writing and Integration
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